Gender Classification using Twitter Text Data

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***Abstract* — Increasingly content sharing websites such as social media have become very popular in many countries across the world. Classifying the gender of a person based on these short messages is an interesting research area that could benefit legal investigation, forensics, marketing analysis, advertising and recommendation. This research will explore the use of Natural Language Processing (NLP) techniques and tweets in a gender classification system. This investigation will compare multiple techniques such as Bag of Words (Term Frequency – Inverse Document Frequency), Word Embedding (W2Vec, GloVe) and traditional Machine Learning techniques (Logistic Regression, Support Vector Machine and Naïve Bayes) in this context.**

A new dataset has been generated to be used as part of this study comprising of the user gender and associated tweets. This dataset was developed due to the unavailability of any public standard dataset with the volume required to perform this investigation. The results have determined that the traditional Bag of Words model did not provide any significant results in classification. However, word embedding models have significantly performed better using multiple machine learning techniques. Therefore, the word embedding models have been proven to be the most effective technique in classifying gender based on twitter text data.

Keywords — Natural Language Processing, Gender Classification, Machine Learning, Twitter, Word Embedding.

I. Introduction

Twitter is one of the most famous social networking websites which allows online users to read and post 280-character messages on their website [1]. These messages are generally referred to as tweets. The message size restriction is one of the reasons Twitter has become very popular among social media users. Twitter provides an equal opportunity to global online users to connect with other users including top celebrities, politicians, famous people, etc. and follow them regularly via reading their tweets. The increasing popularity of social media has created a unique opportunity to learn about human society on a large scale [2]. According to Omnicore agency, which is one of the well-known marketing websites, monthly active twitter use reached more than 330 million users, users tweet more than 500 million tweets per day and about 139 million people actively use Twitter on a daily basis. It has also been discovered that approximately 34% of Twitter users are females and more than 66% are male [3], [4]. Twitter has a default option to display the tweets publicly, however, users can put restrictions on messages so they can only be viewed by approved followers. Twitter's main ideology is focused on sharing tweets publicly. According to Kvamme, over 90% of Twitter users prefer to make their interactions publicly accessible [5].

With increasing use of shared content on social media, classifying gender is gaining significant interest from marketing companies and government. Extracting knowledge from this hidden content could be very beneficial to various organisations for different reasons such as forecasting cinema box office outcome, election prediction, forensics, cybercrime and providing recommendation [5]–[7]. Many users on social media prefer to hide their true identity including user name, age, gender, this is commonly referred to as anonymous profiles. As a result, this requires the development of advanced techniques in identifying the gender of these profiles. In most cases, it's required by forensic experts to track cybercrime [2].

Gender classification can be considered a problem of binary classification, for instance, the goal is to classify the object into two groups based on some features [8]. Public availability of such user text data provides the researcher with ample opportunity to find the patterns within this data. Such unstructured/structured text data can be normalised using different text pre-processing and Natural language processing (NLP) techniques. NLP, which is a subfield of Artificial Intelligence (AI), provide machines the ability to understand spoken/written human language. NLP includes different ways to analyse and understand this information and obtain meaning from human language automatically [9].

This research will utilise multiple Machine Learning (ML) methods to train the classifier by using a labelled twitter dataset to automatically detect the gender of a twitter user by applying different NLP techniques on a tweet text. In general, most NLP tasks, such as sentiment analysis, involve classifying a token sequence for a document or sentence. Therefore, the proposed methodology is to vectorise the tweet text, extract features using TF-IDF (Term Frequency – Inverse Document Frequency) then apply ML classification to train and predict the output label. This will then be tested to measure the performance of the model. As no such labelled twitter data is available at the scale required, this study will involve building the dataset initially which will be available for the public for further research.

II. Research Objectives

Building the dataset is the first objective of this research. Kaggle provides a labelled dataset of about twenty thousand users [10]. This dataset will be used as a bootstrap, to fetch tweets for each of those users and generate a labelled dataset for this study.

The main motivation of this research is to detect gender classification by applying several ML techniques and NLP approaches based on the text features to differentiate between gender characteristics. This types of research has resulted in the development of various applications within the applied research area, for example, psychological analysis and marketing [11]. If an anonymous Twitter users communicates as part of some suspicious activity, it is critical to understand the linguistic profile of the user through analysing their past tweets. This analysis could provide a pattern and help in understanding the background and gender (male or female) of the user [12]. In terms of marketing and recommendation,

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multiple organisations use Twitter to deliver digital adverts for their target gender so they can attract their attention. This has proven to be very effective especially for the e-commerce industry when it comes to increased revenue and reach. Moreover, social networking websites such as Twitter, have a large number of bots which are known to spread fake news and misleading information to the general public, this can result in influencing elections and large scale campaigns. Therefore, it is very important to have the ability to detect and understand the author for these Tweets if the author is human or a bot [13].

III. Gender Classification In Text

Gender classification can be treated as a two-class or binary classification problem in practice for the example given two classes, male or female. The aim of this classification is to assign the anonymous text or message to one of the given class without knowing about the user [14]. It is a very challenging task for machines to analyse text but very easy for humans. The visual inspection often makes it easy and fast for a human to determine the gender class. Language sentence or text messages are the input for these systems. In general, classifying a sequence of tokens, for instance, a document or sentence is the typical NLP machine learning task, i.e., function approximating:

$$f1 \rightarrow (1,0)$$

In this example domain, sentiment, etc. might determine f1. The goal is to categorise the data points into one of the categories as male or female by 1 (male) and 0 (female). However, the only difference here is that it deals with different text pairs in different tasks.

Other features of gender classification must be observed during this classification process such as gender identification and sex. According to the American Psychologist Association sex refers to a biological human state and is normally classified as male, female or intersex. Gender Expression is referred to the interaction and communication of the person in a given culture which can be determined by the way the person's interest in clothing or communication pattern [15]. At present gender classification research has been conducted in the form of sex rather than gender expression. Two possible genders male and female are used for consistency purposes. Hence, gender classification research can be considered as the task of identifying if the Twitter user is male or female. This is completed through analysing their profile content features and the gender expression displayed. Also, some research has focused on investigating and analysing meta-attributes or meaningful important attributes for gender identification [2].

IV. Related Work

The issue with gender detection using twitter data has not previously been researched at scale. Although, similar NLP research has been conducted to automatically detect the gender of a writer of text, such as M. Koppel, 2002 & 2009 and S. Goswami, 2009 & 2012 [16]–[19]. Advancement in ML and text classification techniques has helped the growth of research within this social media domain. This research will examine multiple ML techniques such as Modified Balance Winnow, SVM, Naïve Bayes which have been used in other research projects for classifying gender.

One of the first research studies on detecting gender classification by using latent user attributes was introduced by Rao [20]. The research focused on detecting gender, age, political orientation and nationality. Using status updates, communication behaviour and social network structure, the author aimed to research the usefulness of language content processing. The aim was to discover latent user attributes to automatically detect some of the latent features in text. Rao completed detailed research into investigating what the most important features are and which perform well by applying them to a supervised ML problem [20]. SVM classification technique was applied to conclude the analysis. The research suggested that compared to other social media Twitter provided different socio-linguistic indicators. For example, female users were most likely to use question marks (?), combined exclamation marks (!) and emoticons such as <3, were strong signals. Where LMAO laughing indicators were mostly used by males and LOL by a female. As well as excitements marks, for example, OMG was found to be used by females. Another strong indicator of men tweet was the possessive word features for example "my_gf", "my_wife" and "my_husband" and "my_bf" high indicator for females [20]. A similar finding was observed by Clay while using Nigerian tweets to detect the gender of Twitter users. This research applied a supervised ML approach and used only unigram features from tweets which were stronger indicators to detect the gender of the users. Clay concluded that 'tweets' text data is highly predictive of gender on Twitter [21].

Burger achieved interesting results showing that gender classification is not very effective with profile elements when applying supervised ML approaches for tweet classification. Therefore instead, this research links the users with a URL in their profile field which makes it easier to arrive at the user's blogs and their online media. The author generated 18.5M Twitter users with 213M tweets labelled with gender [22]. Word and character n-grams were then restricted to features of the Tweet content and the profile field which were user name, screen name, and description. An accuracy of 92% was achieved once the profile feature data was linked with Tweets' text by applying balanced winnow2 and SVM classification techniques. The researcher suggested that updating status was most likely completed by females and there were strong gender indicators from females compared to men. Moreover, this research concluded that predicting gender classification using the full name was achieving higher accuracy compared to description and screen names. However, it was argued that tweets performed better in predicting gender since the full name field proved to be very informative. Zamal's research suggested using the features based on the principle of homophily. In other words, feature data of a Twitter profile and the associated content can be used to infer user attributes based on the immediate neighbour's attributes. The research achieved 80.2% accuracy of their prediction model on neighbourhood data and 79.5% only when utilising user data by applying the SVM classifier. This improvement was not substantial but notable [23].

Deitrick researched detecting gender classification by using neural network techniques. This research used a manually labelled dataset of 3,031 tweets. Both, balanced winnow and modified winnow techniques were evaluated. The research achieved 98.5% accuracy by applying both models and 53 ngram features selection [24]. Continuing this research, Miller suggested using ngrams and stream techniques. The study created a labelled dataset of 3,000 users and kept one tweet for each user. Naïve Bayes, Perceptron and word ngram was applied during this experiment. This model

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achieved significant results and achieved 99.3% accuracy by using the Perceptron while tweets length was at least 75 characters [25].

Relationships between linguistic style and gender were suggested in another research study by Bamman on gender detection. A 14,000 tweet corpus was built by clustering user feeds to investigate what relationships exist between language and gender. The experiment achieved 88.0% accuracy by applying an LR classifier. Like Zamal's research, similar accuracy was achieved when experimenting with the gender homophily. However, it didn't have any significant effect on increasing the minimum's model accuracy. The author was not in favour of using the information to detect Twitter users' gender as a binary variable. This research performed investigation and found that Twitter user language was not influenced by the user's network of friends [29].

Some of the recent research suggested there are other features to be considered to perform gender classification [29]. The relationship between gender, social network, and linguistic style was researched and achieved 88% accuracy [30]. Another similar research study achieved 82% of accuracy by using the tweet content features of celebrities by applying the SVM classifier. It is notable that the accuracy increased up to 86% when suggested features of the celebrities were combined. Research by Merler suggested the technique to extract the user attribute from the pictures uploaded on Twitter. Tweets included the visual information that was used to build a 10,000 tweet manual labelled dataset. 76% accuracy was achieved by applying the visual classifier with the semantic content of the pictures. The accuracy increased from 85% to 88% once adding their extra-textual classifier with visual information features [31].

Recent research by Marco Vicente proposed to automatically detect the gender class by using unstructured text details extracted from Twitter's user profile. The author suggested manually defined features to extract useful information from the user's profile attributes such as screen name. Name related features were proposed in this research which contains the user's experience were then evaluated with a 242,000 user dataset. Several supervised ML techniques were evaluated including SVM, LR, Naïve Bayes, K-means and Fuzzy c-Means clustering. A high accuracy of 97% was achieved with a supervised ML classifier to detect Twitter's user gender. About 96% accuracy was also received for this task with the unsupervised approach based on Fuzzy c-Means. The author noticed the advantage of using a labelled training set and the high possibility of better accuracy with the larger dataset [32].

In summary, no twitter gender dataset is available for the research community except the dataset released by Liu and Ruth [26]. This 10,000 user dataset can be used for further research to find better ways of detecting gender classification using text/tweet data. Most of the researchers in this survey focused on gender inference using features data of twitter users such as user name, social network, neighbour attributes, a combination of visual and textual data, profile information clues, etc. Moreover, recent state of the art research by Monte Vicente's proposed the idea of having labelled dataset containing the textual content which can be used for gender inference. There is currently no such public dataset that is available which includes the tweets text data.

V. Methodology

The methodology of this research involved multiple steps including data gathering (building the dataset), data pre-processing, data transformation, data mining and utilising machine learning models.

A. Data Pre-Processing

The data pre-processing is a very critical step because better results can only be achieved with good quality of data. This is achieved by following steps such as feature engineering, visualisations of data and ML classifiers. Since the social media data is unstructured or in other words it's raw and very noisy it requires robust cleaning. The main objective of this step is to remove noisy and inconsistent tweet data. Tweets that carry very little weighting in text context, for example numbers, special character, punctuations, hashtags, emojis, smileys, extra blank space, etc. need to be removed. This step also involves filtering the tweet length required for feature engineering. Once the raw data is filtered and cleaned then we need to ensure tweets are written in the English language, therefore further pre-processing was completed. Finally removing stop words by using NLTK library during run time [33].

In order to ensure the tweets are of a good quality, the tweet pre-processor library has been utilised for cleaning tweets. This library helps in removing unwanted text from tweets: URL's, Hashtag, Mentions, Reserved words, Emojis and Smileys. For example, tweets with URL's wouldn't provide any strong indication of the gender class. Hashtag (#) is often used by users for trending topics. The hashtag might contain unique features required to perform gender classification however since this research only focused on text features to extract from each gender, they have been removed. The next critical step is to ensure the tweet length is reasonable for each user, so further analysis is performed and the length is split into 5-50 words. This filtering will ensure reasonable length is taken into consideration for further steps. Finally, since the research is only based on English tweets, the dataset was further filtered so only tweets in "English" language remain. The 'Spacy' library was very helpful in this operation and it took a few hours to extract the English tweets.

B. Data Transformation

After the cleaning process, the next step is to transform the word sequences into numerical features. This is also a very important task since ML classifiers are not intelligent enough to process text in raw form. The basis for classification is on the principal of instance optimisation, math, statistics, etc. Especially when the raw data is in natural language text format or has high dimensionality added to it, moreover, such data is unstructured, noisy and distributed. In order to achieve the research objective of analysing tweets from a defined dataset and improve the accuracy of gender classification in unstructured text, feature extraction plays a vital role in most cases [34]. TF-IDF is a well known technique to convert the textual representation of information into the Vector Space Model (VSM). This is a very simple, effective and widely used technique in NLP.

C. Data Mining

Once data transformation has completed with multi-dimension features, the next step is to classify the updated data on ML models. A supervised ML approach is considered in this research to predict the output label class correctly as per

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previous research [29], [35], [36]. The following classifiers were considered in this research; LR, MLP, SVM, Naïve Bayes, RF and XGBoost. These are simple yet effective algorithms for binary classification. A base-line is created using a simple pipeline consisting of the TF-IDF and LR model. Further various algorithms are combined with different feature engineering techniques (Word Embeddings – W2Vec and GloVe) and were tested to compare the results with the baseline.

D. Utilising Machine Learning Models

This research utilised six different ML classification algorithms (LR, MLP, SVM, Naïve Bayes, RF, and XGBoost) to predict the gender class on a featured Twitter dataset. SVM, Naïve Bayes, LR have proven to be very successful and widely used in the past with binary classification problems. These techniques are also well-known for good predictive performance regardless the amount of data [21], [26], [29], [31], [32]. Naïve Bayes is known for its simplicity and ease of implementation. This technique has also outperformed alternatives in real applications when it comes to reliability, easy and efficient classification [37]. Similarly, XG boost has the best model performance and execution speed. Additional techniques such as RF and MLP are also tested for this research due to comparison purposes to analyse different result achieved when compared with other well-known classifiers.

VI. Twitter Dataset

This research aims to build a labelled twitter dataset that will be used for gender inference based on tweet text. There is currently no public dataset that contains gender and tweets; therefore, it has been decided that as part of this research a dataset will be developed which will include the gender and their associated tweet data. To initiate this process, a twitter dataset was used having 20,000 rows [10]. The purpose of this dataset was to find twitter 'user ids' and their associated gender.

The research goal is to perform gender classification using text data therefore the only fields required are the user id and gender columns for further processing, therefore data is filtered with two columns. Also, the focus of this research is on gender class male or female because unisex and brand name can not be classified as male or female. To ensure that, two filters are applied to the 'gender:confidence' column. First, 'gender:confidence' column filtered with "1" for both classes which represent 100% gender identification. Secondly, the 'gender:confidence' column filtered with "male" and "female" options. This reduced the dataset from 20,000 to 10,020 users, where 4,653 are male and 5,367 were females.

The Tweets dataset has a lot of noise data that needs to be removed for example; URLs, Hashtag, Mentions, Emojis, Reserved words (RT, FAV), Smileys, Numbers, etc. 'Twitter Pre-Processor' library was used for cleaning tweets. Since the presence of these characters decreases the quality of the corpus built, cleaning tweets was one of the most important steps. Another major step in data pre-processing was to ensure that English tweets were kept for analysis, therefore, the language detection technique was applied using the spacy library. Finally, the NLTK package is used to remove the stopwords at run time. After the cleaning procedure, the data is ready for more feature engineering such as ML classifications [33].

Once noisy data was cleaned, out of 346,000 total tweets, 'greater than 20' tweets were reduced to 41,932. The male tweets left to 19,387 which is almost half of the 41,000 which shows it is a balanced dataset. After language detection, tweets were further reduced. In total, there are approximately 21,021 total tweets from male and female which means almost 50% of the tweet data was in another language.

A. Dimensionality Reduction

Once the pre-processed step is completed, there are approx. 21,021 tweets (male & female). To plot the data dimensionality reduction techniques were used. A number of features were set to two so that it can be plotted on the X-Y axis. The purpose of plotting data is to get an insight around how separable the two classes are. If the two classes are separable via a linear or no-linear boundary, then it should be easy to train a model to achieve desirable performance. Conversely, if the classes are not easily separable, then it could be difficult to achieve desirable performance using simple techniques.

PCA (Principal Component Analysis) works by eliminating the dimensions which contain the least amount of variance between the two classes. Even though the number of features is greatly reduced, PCA ensures that the most important information in data is captured along the reduced number of dimensions.

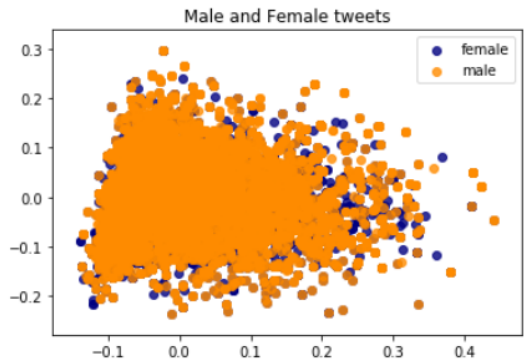

Fig. 1. Male and Female Tweets Plot using PCA

The graph (Fig. 1) shows the data for both genders is overlapping therefore, it's difficult to find if some unique words are used by males and females individually. In other words, there is not much variance in the feature data among male and female tweets.

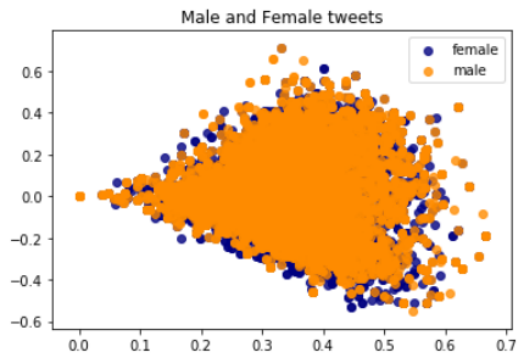

Fig. 2. Male and Female Tweets Plot using truncated SVD

The Truncated SVD (Singular Value Decomposition) technique computes the linear dimensionality through means

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of truncated SVD like PCA but works on sample vectors directly in place of finding covariance matrices. The feature transformed data was then plotted where X and Y represents Male and Female tweets. The graph (Fig. 2) is very similar to PCA due to both maintaining the largest amount of variance.

VII. Feature Generation

A. TF-IDF

Once the data is pre-processed then the next step will be to apply the feature extraction technique on to the updated dataset. Feature extraction techniques help the ML algorithms to understand the text by converting it into numbers. TF-IDF is known as one of the most efficient statistical weighted measured approaches used for feature extraction and is provided in the Scikit-learn library.

B. Word Embedding

The word embedding (W2Vec. GloVe) techniques are expected to detect extra semantic features and reduce dimensions in detecting the gender class. Previous research is mostly involved without traditional BOW approaches e.g. BOW, TF-IDF, N-grams, etc. Therefore, the approach of this research is representing text through the VSM which is more semantically rich than simple TF-IDF.

W2Vec and GloVe models were used as part of the genism library. This research will use a pre-trained vector as these models are stronger regarding word semantics [38] and are trained on a large corpus such as Wikipedia. W2Vec group the vector of similar words in vector space for every unique word which helps in detecting mathematical similarity of word features without any human intervention. Fortunately, genism provides a pre-trained vector on the Google News dataset (approximately 100 billion words) for W2Vec. The model contains 300-dimensional vectors for 3 million words and phrases. Similarly, genism provides GloVe's pre-trained model on 2 billion tweets, 27 billion tokens and 1.2 million pieces of vocabulary [39].

For this experiment, the mean of all word vectors in the tweet has been taken since advanced Deep Learning models were not applied. The length of the resultant vector is the same for W2vec (Google) 300 and GloVe (Twitter) model 200. These vectors are very good at capturing the semantics and even analogies between different words. The same process is repeated to obtain the vector representation for all tweets data.

VIII. Results

Initially, the main dataset is split with train and test sets in an 80:20 ratio. The training set contains 16,816 tweets data while the testing set has 4,205 tweets. Training data is used to fit the model to it and then evaluated with totally unseen data to get the evaluation metrics as little bias as possible. Afterwards, the baseline TF-IDF feature extraction technique is applied to convert the text data into numeric data then fed to the ML classifier to predict the gender class. Finally, word embedding features are utilised to train ML models with more semantic rich representations. Since Naïve Bayes and MLP models do not accept the non-negative values created in word embedding pre-trained models, it is not possible to use these models, and hence were removed from the results.

Table I shows the summary of all the experiments including baseline and Word embeddings (W2Vec, GloVe). The table shows good improvement in the accuracy score once the word embedding model has been introduced with the ML classifiers. The LR model achieved the highest accuracy score of 57.14% whereas the baseline approach only had 53.65% accuracy, an improvement of 6.5% increase.

TABLE I. Comparing Multiple ML Techniques

Final Results	
ML Model	***Accuracy***
Baseline (TF-IDF)	
LR (Logistic Regression)	53.65
MLP (Multilayer Perceptron)	48.75
SVM (Support Vector Machine)	52.67
Naïve Bayes	53.84
Random Forest	47.7
XGBoost	54.93
Word Embedding (W2Vec)	
LR (Logistic Regression)	57.14
SVM (Support Vector Machine)	52.67
Random Forest	47.72
XGBoost	55.38
Word Embedding (GloVe)	
LR (Logistic Regression)	54.43
SVM (Support Vector Machine)	52.67
Random Forest	48.46
XGBoost	52.39

In the case of word embedding models, the performance metrics for the majority of models are improved, the LR model achieved much better results with W2Vec vectors. It also proves that word embedding techniques can detect the semantic meaning, an analogy for different words that possibly helped in improving the performance metrics.

Finally, since our dataset is a unique, random, unbiased, diverse and good representation of social media, this study did not get the same results as compared to the results achieved in similar research completed in past. The dataset curated in this research is truly random and unbiased in a way that it is not curated from any specific groups like celebrities, politicians, etc. The datasets used contain millions of data points that should impact the accuracy of trained models. Regarding getting high accuracy with fewer tweets, it was observed that Miller's [25] research with gender prediction involved manually labelling data as male or female tweets.

IX. Conclusion

This research focused on finding a better system that could provide an autonomous solution of detecting gender class (male or female) by using only the tweet's text data. Initially, the study explores how text data is increasing day by day and the role of social media in generating billions of terabytes every single day. Twitter is one of the most popular social media platforms that generate a lot of text data and received a lot of research attention with such data as a means of understanding, predicting real-life phenomena, monitoring, etc. Also, it discussed the importance of such data for several organisations. Moreover, a brief discussion of the research

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problem is deliberated rearding how text data can be challenging to classify if it's from male or female when it comes from different domains and unstructured format.

The review of related literature including traditional BOW algorithms such as N-grams, TF, TF-IDF, etc. are discussed along with techniques that can help in solving an NLP problem with text classification. Recent research on gender classification using different techniques such as Word ngram, Naïve Bayes, LR, SVM, Perceptron are discussed as well as the impact these studies have on NLP. Furthermore, the needs and importance of NLP for detecting gender classification are discussed to meet the research objective. Dimension reduction techniques (SVD, PCA) also helped to understand the data used to predict the gender class. Finally, the challenges are drawn for the associated main applications. The importance of word embedding is also noticed and realised for feature extractions which can help in extracting semantic meaning from text data and improving model accuracy.

Based on the advantages and disadvantages of recent research, the predictive ML models (LR, MLP, SVM, Naïve Bayes, RF, and XGBoost) are tested on the test set with a model trained on training data. These models are tested with simple TF-IDF and word embedding (W2Vec, GloVe) features to detect the gender classification from tweet text. It is a different approach as compared to the previous research which is more focused on traditional feature extraction techniques such as BOW, TF-IDF, etc. The traditional BOW technique is only focused on counting the word frequency in a document. The best predictive model (W2Vec + LR) helped in optimising the accuracy matrices where the baseline model (simple TF-IDF + LR) is outperformed. Based on the results of this research, it is proven that classifying gender is more predictable with advanced feature extraction techniques rather than using traditional BOW approaches that do not take semantic meaning, analogy and context around the nearby words into consideration.

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